

Summary

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Background:

The Aquatic Intermittency Effects on Microbiomes (AIMS) project is a multi-agency collaborative research effort with the goal to understand how stream intermittency (periods of drying and rewetting) influences stream ecosystems. Within the project, in-stream sensors are deployed at monitoring sites throughout multiple watersheds to collect Stream Temperature, Intermittency, and Conductivity (STIC) data. This data consists of time series observations from different sites across the United States. Each site conducts a measurement every 15 minutes, collecting the time, device used, conductivity (uS/cm), Temperature (C), stream status (1 = wet, 0 = dry), and a read quality flag (Poor-Good). There are three regions included within the STIC data, each containing three watersheds. These regions span a broad range of climatic and hydrologic conditions. Regions included are:

Great Plains: *Kings Creek - Konza Prairie Biological Station* *Shane Creek - Konza Prairie Biological Station* *Young Meyer Ranch - Colorado* Mountain West: *Johnston Draw - Idaho* *Gibson Jack Creek - Idaho* *Dry Creek - Idaho*. South East: *Talladega - Alabama* *Paint Rock - Alabama* *Shambley Creek - Alabama*.

The Raw STIC data contains 14 columns and approximately 22 million rows of observations. Columns include the project (project), date and time (datetime), the unique site ID (siteId), the resource type (rtype), the sub location of the instrument (sublocation), the serial number of the device (SN), the conductivity (condUncal), Temperature (TempC), deployment period (rep), specific conductivity (SpC), whether the measurement was wet or dry (wetdry), the quality of the measurement (qual_rating), an internal code for the quality (QAQC), and the Year the measurement took place (year).

Research Question:

Which monitoring sites and watersheds exhibited the highest annual proportion of dry days, and how does watershed intermittency change by year?

Methods:

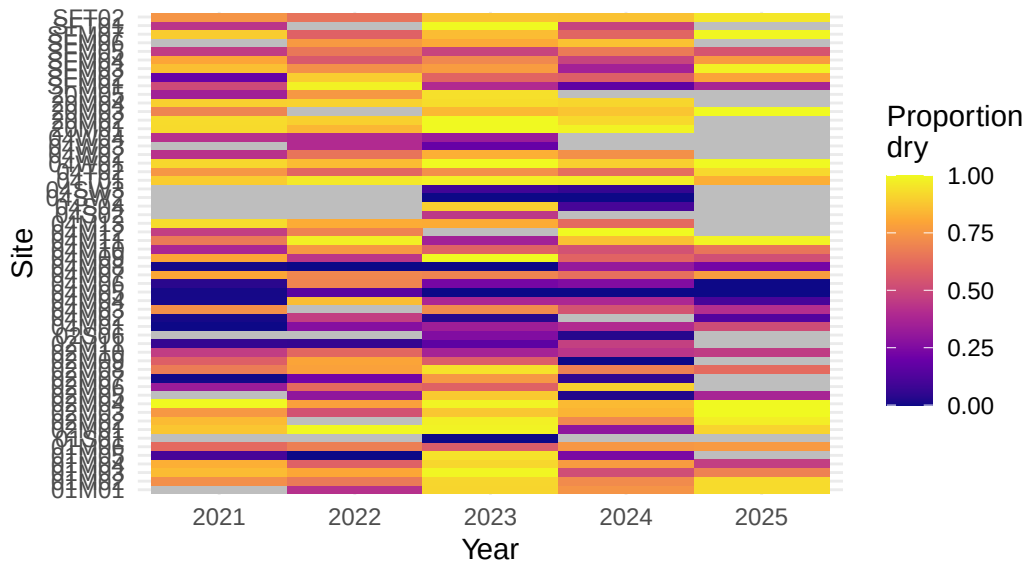
Raw files were obtained from Hydroshare and unzipped. Extracted files were put into a directory named “extracted.” If no such directory exists, one will be created. Because the downloaded archives contained multiple directory structures and file formats, extraction required a custom workflow. CSVs were selected for loading using a grep queries for “.csv.” Duplicate files were identified using MD5 hash values and removed prior to import. The data frame was split into individual watersheds and checked to confirm only the data requested was transferred. A correctly functioning check will return “True” and “0”. The watersheds within the same region were then concatenated into region by region data frames. Using similar methods as Plont et al. (2026)[1], low quality measurements were discarded by looking at the proportion of poor reads per serial number per year. If the proportion of poor reads were greater than 50% for a given year, that serial number was not included in analysis. Because monitoring duration differed among site-years, annual summaries were based on the proportion of observed days classified as dry rather than the total number of dry days. This standardization improves comparability among sites with differing monitoring duration. Annual observation counts were summarized separately to document differences in monitoring duration among site-years. The quality controlled data was then used to compute yearly summaries per site. These yearly summaries give the proportion of yearly days that were classified as dry. A day was classified as dry when more than 50% of valid observations recorded that day indicated dry conditions; otherwise the day was classified as wet. Averages were taken, ranked in descending order, and displayed for the tables. For ranking tables only, site-years with observation totals more than one standard deviation below the regional mean were excluded. This filtering reduced the influence of sites with limited monitoring duration on estimates of average annual dryness. All heat maps retained the complete quality-controlled data set.

Deliverables:

Example Dry Day Proportion Heatmap:

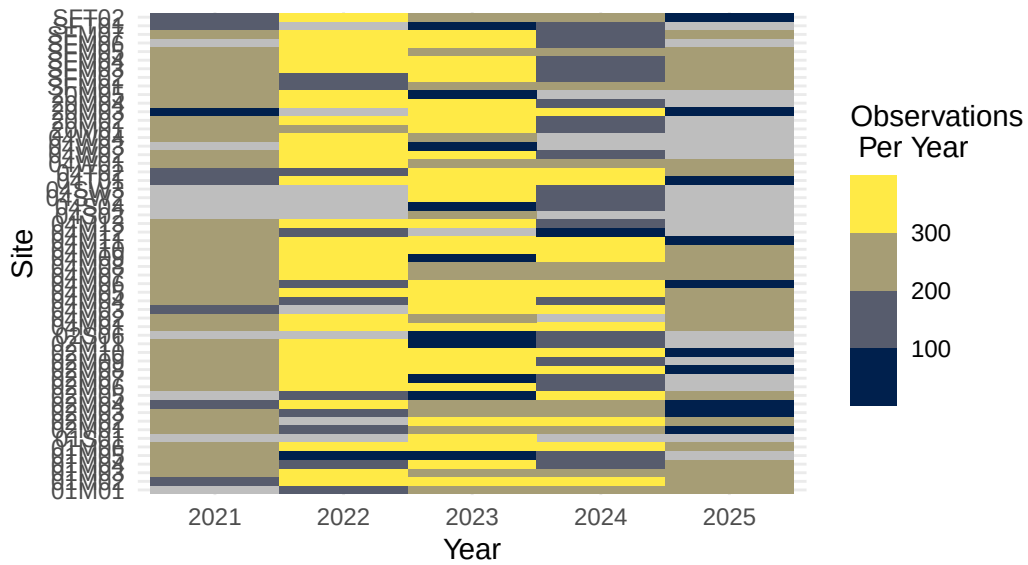
``summarise()`` has grouped output by 'Date', 'siteId', 'Year'. You can override using the ``groups`` argument.

Annual Proportion of Dry Days at Kings Creek
Gray values were unobserved



Example Data Coverage/Observation Heatmap

Number of Included Observations at Kings Creek Konza Pra
Light Gray values were unobserved



Supplementary Tables:

Site by Site Examination of Dry Day Proportions:

Table 1: Great Plains Sites with Highest Proportion of Dry Days

Site	Watershed	Average Proportion of Dry Days
SHM03	SHN	0.9961
04T01	KNZ	0.9670
SHA04	SHN	0.9648
20M02	KNZ	0.9429
20M01	KNZ	0.9331
04W01	KNZ	0.9316

Watershed Average Dry Day Proportions Table:

Table 2: Proportion of Dry Days by Watershed

Watershed	Average Proportion of Dry Days
KNZ	0.5649
SHN	0.6935
YMR	0.6162

Conclusions:

Using average dry-day proportions by watershed, the Great Plains emerged as the driest region overall. All three of its watersheds had higher dry-day proportions ((0.5649), (0.6935), (0.6162)) than even the driest streams in the South East (PRF - 0.4625) or Mountain West (JDR - 0.4663). The heat maps support this, showing very high levels of dry days and only certain sites measuring consistent hydration year to year. No Great Plains watershed measured a dry day proportion lower than 0.55, meaning that more often than not sensors recorded dry days.

There are differences in Intermittency within years within watersheds. The watershed heatmaps illustrate important differences between years. Dry proportions can be used to see the hydration for a given year and differences between years in a watershed. This can especially be seen in the Great Plains region, where 2023 proved to be drier universally among Great Plains watersheds than years following or preceding.

Perhaps more importantly, these heat maps show the gaps in current sensor information. Filtering removed many sensors for poor quality, leaving some areas drastically lacking in coverage. While the data does showcase the Great Plains region as the driest, Shane Creek

and Gibson Jack Creek show significant gaps in coverage. Furthermore, the South East region showcases a need for greater sensor quality and quantity deployments. While some sensors were removed by the quality control process, addition and subtraction of certain sites throughout the study period lead to more substantial losses.

As it stands, looking at why sensors malfunction will lead to greater coverage in the future. This can be coupled with rigorous testing of sensor read quality to address data gaps should a certain instrument be malfunctioning. Furthermore, standardizing deployment time of sensors will lead to more accurate conclusions, and ability to count dry days for a cohesive look at each watershed. Further studies into dry day totals could be supported with standardized deployment timing, and equal coverage of sensors across a watershed.